CADRE

Coordinated Access for Data, Research and Environments

Dr Ryan Perry

Director, Australian Data Archive

Partners

Universities

Government agencies

eResearch Providers

Motivation and aims

Why CADRE?

- Challenges: Sensitive data access is inconsistent, inefficient. Need to comply with recent legislation (DAT Act 2022).
- The solution: CADRE provides a shared, flexible platform to simplify access management while ensuring robust security.
- **What We'll Cover Today:**

1. How CADRE operationalises the Five Safes framework.
2. Key features and demonstrations of data user and data provider dashboards.
3. Future plans for CADRE and how you can get involved

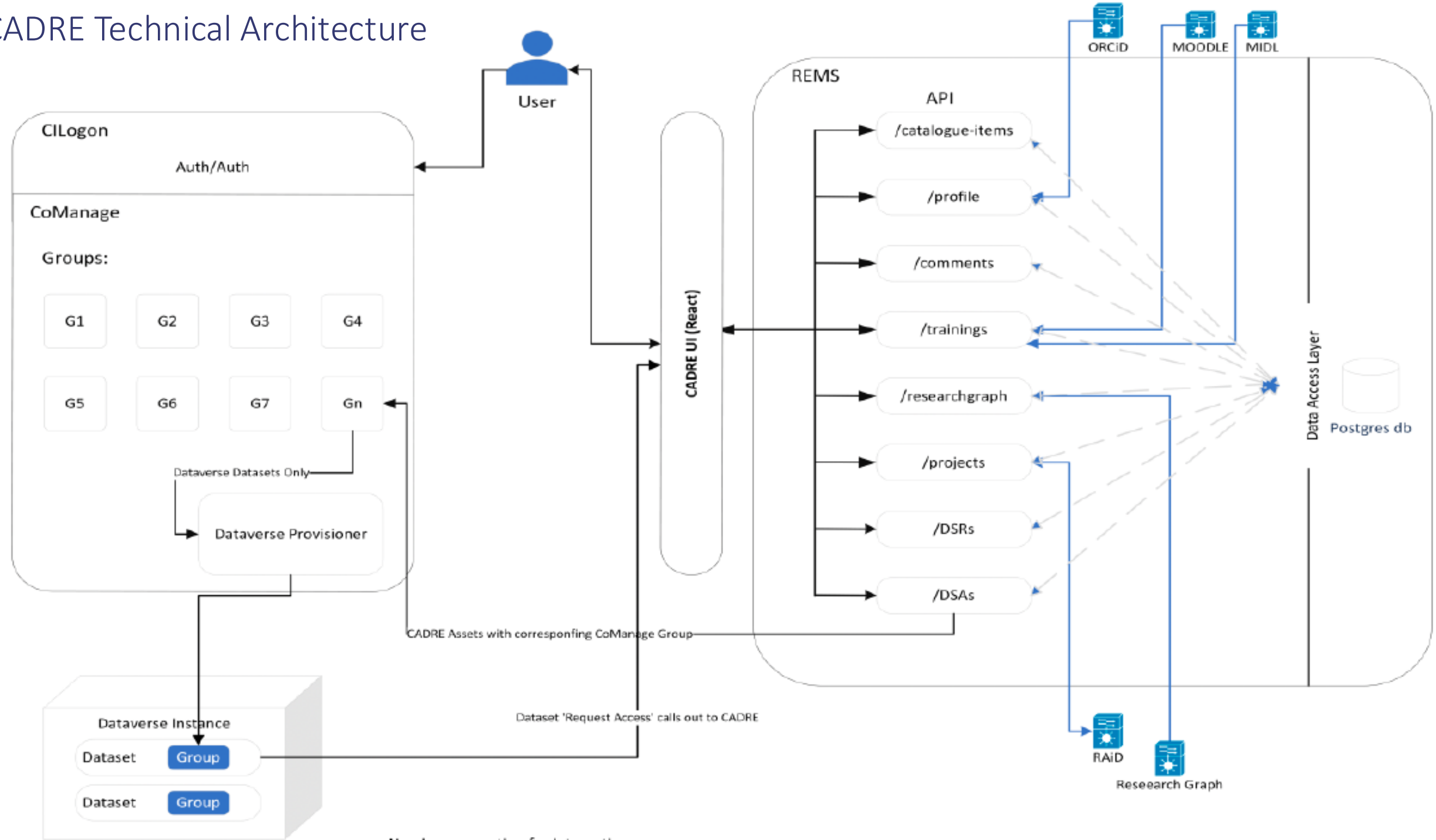
What are the five safes?

Safe dimension	**2016 conception (Desai, Ritchie & Welpton)**	**2022 ONDC Data Sharing Principles**
Safe projects	Is this use of the data appropriate?	Data is shared for an appropriate purpose that delivers a public benefit
Safe people	Can the researchers be trusted to use it in an appropriate manner?	The user has the appropriate authority to access the data
Safe data	Is there a disclosure risk in the data itself?	Appropriate and proportionate protections are applied to the data
Safe settings	Does the access facility limit unauthorised us?	The environment in which the data is shared minimises the risk of unauthorised use or disclosure
Safe outputs	Are the statistical results non-disclosive?	The output from the data sharing arrangement is appropriately safeguarded before any further sharing or release

Implementing five-safes

Data sharing – Fragmented/Ad hoc	Data sharing - CADRE
Data users (people) complete one-off, ad hoc applications using **standalone, manual** processes	Users can access, link and reuse information from their projects, profile and collaborators in a coordinated and integrated way
Data custodians (people) must make stand-alone decisions **in isolation** from other data, projects or people	Projects, people and settings are visible and auditable by custodians in evaluation of applications
To conduct ***research projects,*** users and custodians must manage multiple applications for multiple datasets with multiple custodians	Projects can integrate people, projects and data into a coordinated application
Secure access facilities (settings) set up individual, one-off identities independent of institutional or national identity services	Identities are established and managed through single sign-on services leveraging (inter)national identity providers (AAF, CILogon)
Outputs of sensitive data projects are disconnected from how they were approved or produced	Outputs can potentially be tied to the projects, people and settings that generated them

CADRE Technical Architecture



Key features

- Authentication
- Coordinated requests
- Training materials
- Data sharing agreements

Data Users

- Integrated data providers
- Custom workflows

Data Providers

Authentication with CILogon

- Open-source identity and access management platform for research collaborations
- Federated identity management
  - uses your organisation identity to access other applications
  - supports over 5000 identity providers
- Collaborative organisation management
  - enables the defining of user groups for authorisation
  - customised enrolment flows & identity matching
  - provisioning of access to data and environments

CADRE

AUSTRALIAN ACCESS FEDERATION AAF.EDU.AU

Consent to Attribute Release

stage.cadre.ada.edu.au requests access to the following information. If you do not approve this request, do not proceed.

- Your CILogon user identifier
- Your name
- Your email address
- Your username and affiliation from your identity provider

Choose Your Identity Provider

Australian National University

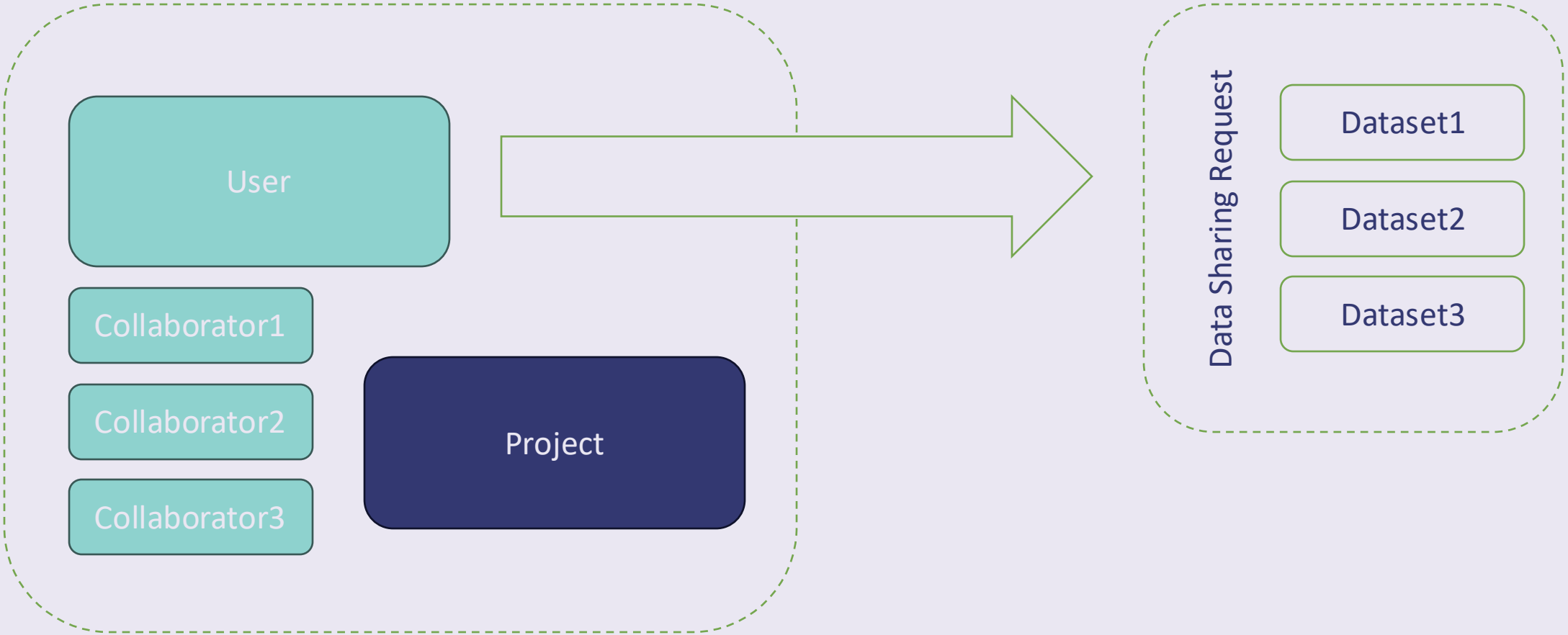
Remember this selection

LOG ON

By selecting "Log On", you agree to the privacy policy.

Can't find your login server? Click here to create a new CADRE Guest account

Coordinated requests



Safe project assessment

- Safe project assessment may take priority over the other four safes (Ritchie and Tava, UWE Bristol, 2020)
- Considers use of data in the context of legislation and ethics
  - Consistent with original purpose of collecting data
  - In the public interest
- Indicators to consider
  - Intended use
  - Context
  - Ethics
  - Risks
  - Collaborators

Training materials: CADRE Learning Site

- Safe people are those who have sufficient experience working with sensitive data. Technical skills was identified as a gap across the research industry.
- Online courses were developed for CADRE
  - Five safes for sensitive data management
  - Roles and responsibilities
- Training course materials can be added to the CADRE Learning site
  - Completion can be set as a requirement for access to data
  - Completion status will appear in a user’s profile on their request

Data sharing agreements

- A shared record of who has access to data assets and under what conditions
- For every request that is initiated, CADRE generates a PDF record of
  - Applicant details
  - Data assets
  - Terms of use
  - Additional application details
  - Event log
- Available to download for applicants and approvers at any stage of the request process

Access management dashboard

Demonstration 1

Integration of data providers

- CADRE is designed as a shared access management platform for multiple data providers, each with distributed storage and provisioning.
- Providers are integrated on CADRE as organization units (COUs)
  - COUs can add resources that belong to their organization and specify workflows
- API access for external data portals
  - E.g., ADA, LDaCA
  - Full functionality for data providers of any size

CADRE workflow management

- Organisation owners set access conditions and approval processes by assigning the following components to a workflow:
  - Resources
  - Approvers
  - Forms
  - Licenses
- Workflows can be configured to automate parts of the approval process and send notifications.
- All actions defined in a workflow are recorded in a log.

CADRE integration process

Demonstration 2

Governance Policies

- CADRE is the first Australian entity to adopt the Authentication and Authorisation for Research Collaborations (AARC) framework
  - Adapted from the AAF NRI Policy Development Toolkit
- The framework is widely used internationally, including:
  - European Open Science Cloud (EOSC-Life)
  - Large Hadron Collider (LCH)
  - National Institutes of Health (NIH)
- Establishes a shared sense of trust for users, providers, and administrators across distributed research services.

AARC

AUSTRALIAN ACCESS FEDERATION

Future development

- Organisation-specific user groups
  - Collaborative Organisations (COs) have more control over policies, workflows and user management but requires a larger investment in administration.
- Service integrations including RAiD

UI Enhancements

- Hidden resources
- Expanded search (while avoiding duplication with discovery services)
- Populating requests: Including existing approvals for safe people evaluation
- Canned responses

All proposed features are subject to feasibility studies

Get in touch to

- Schedule a demo
- Join CADRE to manage your data access

admin@cadre5safes.org.au